HAOYU WANG

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PhD candidate in Financial Engineering. Research interests: machine-learning asset pricing, quantitative investing, portfolio optimization and market microstructure.

Education

University of Auckland / *PhD, Financial Engineering - Full Scholarship* — Dec 2024 - Present

Xi'an Jiaotong-Liverpool University (XJTLU) / *MSc, Financial Mathematics* — Sep 2022 - Apr 2024

Chongqing Finance and Economics University / *BSc, Finance* — Sep 2018 - Jun 2022

Research, Industry & Teaching Experience

University of Auckland / *Financial Researcher* — Dec 2024 - Present

University of Auckland / *Graduate Teaching Assistant, Fintech* — Jul 2025 - Present

Oxford Finance Limited / *Finance Analyst (Contract)* — Oct 2025 - Dec 2025

CUHK, Shenzhen Research Institute / *Research Assistant (Contract)* — Jan 2024 - Dec 2024

Guohua Life Insurance Co., Ltd. / *Actuarial Intern* — Mar 2024 - May 2024

S&P Global / *Quantitative Research Intern, IHS Automotive Planning Solution* — Oct 2023 - Feb 2024

Xi'an Jiaotong-Liverpool University (XJTLU) / *Research Assistant, School of Mathematics and Physics* — Oct 2022 - Feb 2024

Kaiyuan Securities / *Quantitative Intern* — Aug 2023 - Oct 2023

Volvo Group / *Data Analyst Intern* — Mar 2023 - Aug 2023

Research: ML asset pricing, factor discovery, portfolio optimization and market microstructure. S&P Global: ML vehicle-pricing analysis and the 2023 China NEV Export Outlook. Additional teaching: Calculus TA, XJTLU (2023).

Publications

Machine learning predictive uncertainty and stock returns: Evidence from LightGBM entropy in the U.S. market (2026). with Dejun Xie, Wenze Xiong and Jiale Zhang. *Finance Research Letters*, 110955. DOI

Optimizing Stock Market Prediction and Stock Trading Strategies with Deep Learning Models Enhanced by Nonlinear Feature Identification and Robust Prediction Evaluation (2026). *Financial Innovation*, 12, 117. DOI

Information-aware Valuation and Dynamic Pricing for Textual Data in Mobile Environments (2026). *Information Processing and Management*, 63 (7), 104855. DOI

Can Ensemble Learning Reveal Real Estate Value – A Study of England and Wales Cases Anonymous (2026). with Yinze Ji and Yishuang Xu. *Journal of Chinese Economic and Business Studies*, 1–36. DOI

Optimal Profit-Making Strategies in Stock Market with Algorithmic Trading (2024). *Quantitative Finance and Economics*, 8 (3), 546.

Working Papers

Too Much of a Good Thing: Characteristic Redundancy and Machine-Learning Portfolio Performance — 2026

With D. Jayasuriya, L. Xiao and X. Han. Working paper; AFBC 2026 main program.

Entropy-Regularized Portfolio Optimization with Reinforcement Learning Control — 2025

Working paper. SSRN 5276183.

Selected Projects

FactorForge - LLM Factor Research - LLM-guided factor discovery, diagnostics and out-of-sample portfolios across US/CN equities and crypto.

Hybrid RL + Avellaneda-Stoikov - RL market making with an Avellaneda-Stoikov baseline.

Avellaneda-Stoikov Market Making - BTCUSDT order-book research and inventory control.

Limit Order Book Replay - Order-book replay tools for microstructure research.

A-share Alpha Factor Lab - Alpha158/GTJA191 factors, walk-forward LightGBM and Tushare data.

Momentum Is All You Need - Skip-month equity momentum, volatility targeting and capacity analysis.

NZ Electricity Price Forecasting - Half-hourly NZEM spot-price forecasting using Electricity Authority/EMI data.

Presentations, Awards & Certifications

39th AFBC, UNSW Sydney, main program (Dec 2026, upcoming); **Citadel Trading Invitational**, Australia (2025).
XJTLU Entrance Scholarship (2022); CMA Part 2 passed - Strategic Financial Management; Securities Practitioner Qualifications, 3 modules (2019); NCRE Level II (2019).